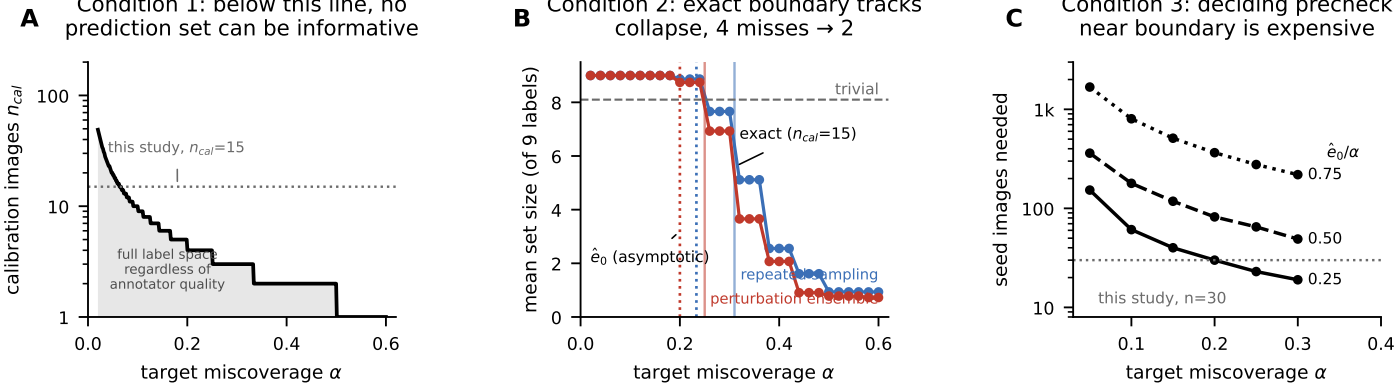
A
Condition 1: below this line, no prediction set can be informative
calibration images n_{cal}
100
10
1
this study, n_{cal}=15
full label space regardless of annotator quality
0.0
0.2
0.4
0.6
target miscoverage α
B
Condition 2: exact boundary tracks collapse, 4 misses → 2
mean set size (of 9 labels)
0
2
4
6
8
trivial
exact (n_{cal}=15)
$\hat{e}_0$ (asymptotic)
repeated sampling
perturbation ensembles
0.0
0.2
0.4
0.6
target miscoverage α
C
Condition 3: deciding precheck near boundary is expensive
seed images needed
1k
100
10
$\hat{e}_0/\alpha$
0.75
0.50
0.25
this study, n=30
0.1
0.2
0.3
0.4
target miscoverage α